SKELETON-GROUNDED LATENT ACTION MODEL FOR ROBOT WORLD MODELS

Qi Lv[1,2] **Jiayi Lin**[1,3] **Yuxin Jiang**[2] **Jianzong Wang**[3]
Xiang Deng[1] **Mike Shou Zheng**[2]✉
[1]HIT (Shenzhen) [2]Show Lab, NUS [3]Ping An Tech

Project page: https://aopolin-lv.github.io/SkeLAM/

ABSTRACT

Robot world models simulate how environments evolve in response to robot actions. Latent action models enable learning dynamics from videos by encoding visual transitions as proxy actions. However, reconstruction-driven latent actions mix robot motion with object responses and other scene changes, complicating their correspondence with robot controls. We propose **SkeLAM**, a skeleton-guided latent action model for controllable robot world modeling. SkeLAM decomposes latent actions into a kinematic component grounded by robot skeleton supervision and a residual component retaining complementary visual dynamics. Cross-modal swap reconstruction encourages RGB-derived and skeleton-derived kinematic representations to be interchangeable during video reconstruction. The world model incorporates both components through separate conditioning pathways, while branch dropout reduces its dependence on future-derived residual information. A lightweight adapter maps robot actions to the kinematic space, and subsequent world-model fine-tuning adapts generation to these predicted conditions. At inference, the model generates future observations from the current observation and robot actions. On DROID, SkeLAM reduces end-effector ADE by 19.5% and LPIPS by 11.1% compared with DreamDojo. Motion probing supports kinematic specialization, while qualitative results demonstrate zero-shot human-to-robot motion transfer in generated rollouts. Policy evaluation shows agreement between predicted success rates and those measured in real robot executions.

1 INTRODUCTION

Robot world models simulate how environments evolve in response to robot actions, supporting policy evaluation and model-based planning (Yang et al., 2024; Wu et al., 2024). Accurate simulation requires modeling robot motion and resulting scene changes. Latent action models encode visual transitions as proxy actions (Bruce et al., 2024; Ye et al., 2025), allowing world models to learn dynamics from large-scale videos without explicit action annotations (Gao et al., 2026). These latent actions are learned through future-observation reconstruction, with the current observation providing visual context (Gao et al., 2025; Liang et al., 2026). For action-conditioned simulation, inferred latent actions serve as supervision for mapping robot controls into the learned latent action space.

Learning this mapping is challenging since reconstruction objectives encourage latent actions to encode robot motion together with object responses and other scene changes (Nikulin et al., 2025; Zhang et al., 2025). For example, similar end-effector motions may produce different object responses under different contact states (Gu et al., 2026; Xu et al., 2026), introducing scene-dependent variation beyond robot motion into the adaptation target. These interaction outcomes are useful for world modeling, since accurate simulation requires predicting how robot motion affects the surrounding scene. This motivates a dedicated kinematic component for action adaptation, together with a residual component retaining complementary transition information for dynamics learning.

To this end, we propose SkeLAM, a skeleton-guided latent action model for controllable robot world modeling. Robot skeletons describe robot configuration and motion while abstracting scene appearance. Related robot world models use skeleton sequences or kinematic renderings as direct conditioning signals for generation (Wang et al., 2025b; Wu & Gao, 2026; Xu et al., 2026). SkeLAM uses paired RGB and skeleton transitions to learn kinematic and residual components from RGB

observations. Skeleton supervision grounds the kinematic latent in robot motion, while the residual latent supplies complementary information for RGB reconstruction. Together, the two components jointly support reconstruction, providing a dedicated target for action adaptation while retaining information for dynamics learning. During representation learning, skeletons supervise the RGB latents, while the deployed interface uses RGB observations and robot actions.

We further introduce cross-modal swap reconstruction, which replaces the RGB kinematic latent with its paired skeleton counterpart while retaining the RGB residual. Reconstructing the same transition under this substitution encourages compatible motion conditioning in the RGB decoder, extending cross-modal correspondence beyond alignment in latent space. The world model learns from both components through separate conditioning pathways, while branch dropout prepares it for the zero-residual condition used at deployment. A lightweight adapter maps robot actions to the kinematic space, and subsequent world-model fine-tuning adapts generation to these predicted conditions. At inference, generation uses the current observation and robot actions.

To assess motion accuracy and visual fidelity, we evaluate SkeLAM on DROID (Khazatsky et al., 2024). Compared with DreamDojo, SkeLAM reduces end-effector ADE by 19.5% and LPIPS by 11.1%. Motion probing shows that the kinematic component approaches the performance of the concatenated representation with half the latent dimensions, while latent interventions and component ablations support its role in generation. Qualitative results further demonstrate zero-shot human-to-robot motion transfer in generated rollouts. Open-loop policy evaluation shows agreement between predicted success rates and those measured in real robot executions.

Our contributions are summarized as follows:

- We propose SkeLAM, a skeleton-guided latent action model that separates the conditioning roles of robot motion and complementary visual dynamics, providing a dedicated kinematic target for robot action adaptation.
- We introduce cross-modal swap reconstruction to encourage compatible RGB and skeleton motion conditioning during generation. Combined with branch dropout, this design supports robot action conditioning through the kinematic pathway learned from visual transitions.
- Experiments on DROID demonstrate gains in motion accuracy and visual fidelity, supported by representation analyses and component ablations. We further show zero-shot human-to-robot motion transfer and agreement between predicted and measured policy success rates.

2 Related Work

Action-conditioned robot world models. Robot world models support interactive simulation (Yang et al., 2024; Wu et al., 2024; Wang et al., 2025a) and policy evaluation (Zhu et al., 2025; Guo et al., 2026; Liao et al., 2025). Visual prediction supports foresight-guided policy learning (Lv et al., 2025a; Cai et al., 2026a) and multi-view data augmentation (Wang et al., 2026). Related research addresses video procedure planning (Ye et al., 2026), temporal answer grounding (Zhang et al., 2026), and cloud–edge agent execution (Huang et al., 2026). DALI-R (Luo et al., 2026) reranks actions with 3D point-cloud latent predictions and task-completion scores. Control representations include action embeddings, skeleton sequences (Wang et al., 2025b; Wu & Gao, 2026), and kinematic renderings (Xu et al., 2026). Recent models further incorporate spatial grounding (Gu et al., 2026; Yang et al., 2026b). Separating camera and object motion also supports camera-controlled video generation (Cheng et al., 2026b).

Latent action learning from videos. Latent action models encode visual transitions for world modeling (Bruce et al., 2024; Gao et al., 2025; 2026; Garrido et al., 2026) and robot policy learning (Ye et al., 2025; Liang et al., 2026; Yang et al., 2026a). Reconstruction-based learning is sensitive to distracting dynamics and ambiguous action correspondence (Nikulin et al., 2025; Zhang et al., 2025). Olaf-World (Jiang et al., 2026) aligns integrated latent actions with temporal differences in frozen video features across contexts. LAOF (Bu et al., 2026b) uses agent optical flow as motion supervision to reduce sensitivity to distractors. Shared RGB–depth features provide geometric supervision (Govind et al., 2026), alongside advances in latent representations for policy learning (Liu et al., 2026a; Shen et al., 2026; Xie et al., 2026; Liu et al., 2026b). LARYBench (Nie et al., 2026) evaluates representations using semantic action classification and physical action regression, independently of downstream policies. A systematic study finds that proxy metrics support coarse model

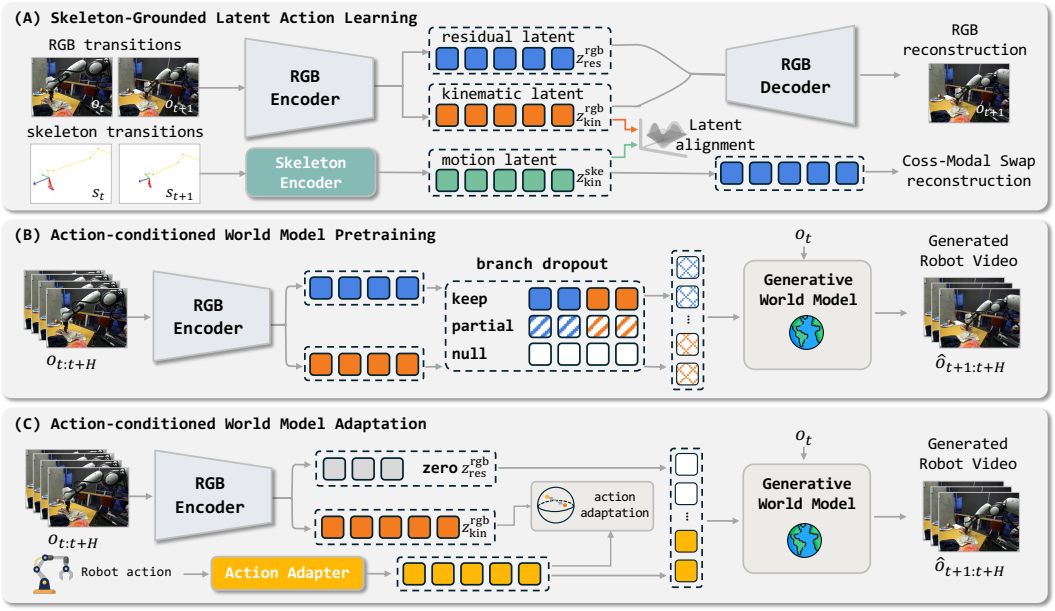


Figure 1: **Overview of SkeLAM.** (A) Skeleton supervision and cross-modal swap reconstruction structure RGB transition latents into kinematic and residual components. (B) Separate projections condition the world model: keep retains both branches, partial drops one, and null drops both. (C) An action adapter predicts kinematic latents, with the null residual at inference.

selection but poorly rank downstream policy performance (Bu et al., 2026a). SkeLAM grounds its kinematic component in robot skeletons while retaining complementary visual dynamics in a separate residual.

Structured action representations. Quantized skills support autoregressive action generation (Li et al., 2025), while flow-matching policies preserve latent-action structure and align latent and physical actions through interpolation-based sampling during decoder training (Lyu et al., 2026). Spatial and kinematic priors include voxel heatmaps (Yang et al., 2026c), robot graphs with differentiable kinematics (Lv et al., 2025b), and trajectories defined by endpoint positions and velocities (Lv et al., 2026). Joint image, LiDAR, and action modeling supports autonomous-driving trajectory generation (Cheng et al., 2026a). SkeleWAM (Sheng et al., 2026) jointly learns action and future-state prediction over sparse 3D robot–object skeletons. Skel-WAM (Cai et al., 2026b) cotrains visual and hand-skeleton dynamics on human and robot videos, with a robot-trained action expert. SkeLAM uses skeleton supervision to ground video-derived kinematic latents for action-conditioned video prediction.

3 Methodology

SkeLAM learns structured latent actions for video-based world modeling and subsequent adaptation to robot controls. As illustrated in Figure 1, skeleton supervision grounds a kinematic component in robot motion, while a residual component retains complementary visual dynamics. The world model learns from both components, with branch dropout preparing it for action-conditioned inference. An action adapter is then trained to predict kinematic latents from robot actions, followed by world-model fine-tuning using these predictions.

Preliminary. Let o_t and a_t denote the visual observation and robot action at time t. An action-conditioned world model represents the distribution of H future observations:

$$p(o_{t+1:t+H} \mid o_t, a_{t:t+H-1}) \,. \tag{1}$$

To learn dynamics from videos, latent action models infer proxy actions z_t from visual transitions (o_t, o_{t+1}). These representations are learned by reconstructing o_{t+1} from o_t and z_t, and subsequently provide conditioning signals for world-model training.

3.1 SKELETON-GROUNDED LATENT ACTIONS

Latent decomposition. To learn a kinematic representation for robot action adaptation while retaining complementary information for RGB reconstruction, we encode each RGB transition into kinematic latents $z^{\text{rgb}}_{\text{kin},t}$ and residual latents $z^{\text{rgb}}_{\text{res},t}$. Further, to supervise the kinematic latent, we use skeleton images depicting the corresponding robot joint configurations.

Given paired RGB observations (o_t, o_{t+1}) and their corresponding skeleton images (s_t, s_{t+1}), Transformer encoders parameterize Gaussian posteriors for the RGB kinematic and residual latents, and the skeleton motion latent:

$$\begin{aligned}\left(z^{\text{rgb}}_{\text{kin},t}, z^{\text{rgb}}_{\text{res},t}\right) &\sim q^{\text{rgb}}_{\phi}(z_{\text{kin}}, z_{\text{res}} \mid o_t, o_{t+1}),\\ z^{\text{ske}}_{\text{kin},t} &\sim q^{\text{ske}}_{\eta}(z_{\text{kin}} \mid s_t, s_{t+1}).\end{aligned} \tag{2}$$

where $z^{\text{rgb}}_{\text{kin},t}, z^{\text{ske}}_{\text{kin},t} \in \mathbb{R}^{d_{\text{kin}}}$ and $z^{\text{rgb}}_{\text{res},t} \in \mathbb{R}^{d_{\text{res}}}$. The d_{kin} and d_{res} denote the dimensions of the kinematic and residual latents, respectively. We initialize the RGB and skeleton branches through conditional reconstruction:

$$\begin{aligned}\hat{o}_{t+1} &= D_{\text{rgb}}\left(o_t, z^{\text{rgb}}_{\text{kin},t}, z^{\text{rgb}}_{\text{res},t}\right),\\ \hat{s}_{t+1} &= D_{\text{ske}}\left(s_t, z^{\text{ske}}_{\text{kin},t}\right).\end{aligned} \tag{3}$$

Here, skeleton reconstruction establishes the motion reference used for subsequent supervision of the RGB kinematic representation.

Skeleton grounding. To ground the RGB kinematic representation in robot motion, we freeze the trained skeleton encoder and decoder. We reconstruct the next skeleton from the RGB kinematic posterior mean through the frozen decoder and align it with the paired skeleton motion representation. Let $u_t = \text{norm}(\mu^{\text{rgb}}_{\text{kin},t})$ and $v_t = \text{sg}[\text{norm}(\mu^{\text{ske}}_{\text{kin},t})]$, where μ is the posterior mean, norm denotes ℓ_2 normalization, and sg denotes stop-gradient. The grounding and alignment objectives are:

$$\begin{aligned}\mathcal{L}_{\text{ground}} &= \mathbb{E}\left[\ell_{\text{ske}}\left(D_{\text{ske}}(s_t, \mu^{\text{rgb}}_{\text{kin},t}), s_{t+1}\right)\right],\\ \mathcal{L}_{\text{align}} &= \mathbb{E}\left[1 - u_t^{\top} v_t + \frac{\|u_t - v_t\|_2^2}{d_{\text{kin}}}\right].\end{aligned} \tag{4}$$

Here, ℓ_{ske} is the skeleton reconstruction loss detailed in Appendix B.1.

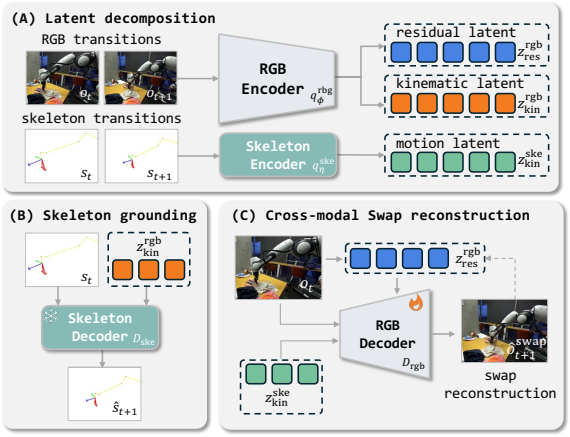


Figure 2: **Skeleton-grounded latent action learning.** (A) RGB and skeleton transition encoding. (B) Grounding through a frozen skeleton decoder. (C) Swap reconstruction using the paired skeleton mean and the same RGB residual sample.

Cross-modal swap reconstruction. To extend motion supervision from the skeleton branch to the RGB decoder, we introduce cross-modal swap reconstruction. We replace the RGB kinematic input with the posterior mean of the paired skeleton motion latent, while retaining the residual sample used in RGB reconstruction:

$$\begin{aligned}\hat{o}^{\text{swap}}_{t+1} &= D_{\text{rgb}}\left(o_t, \text{sg}\left[\mu^{\text{ske}}_{\text{kin},t}\right], z^{\text{rgb}}_{\text{res},t}\right),\\ \mathcal{L}_{\text{swap}} &= \mathbb{E}\left[\text{MAE}\left(\hat{o}^{\text{swap}}_{t+1}, o_{t+1}\right)\right].\end{aligned} \tag{5}$$

The RGB and swapped reconstructions share the decoder, current observation, and target, encouraging compatible motion conditioning across modalities. Gradients propagate to ϕ through the residual latent, while the skeleton branch remains frozen.

Latent-action objective. The RGB objective combines reconstruction MSE with KL regularization of the kinematic and residual posteriors toward standard Gaussian priors:

$$\begin{aligned}\mathcal{L}_{\text{rgb}} = \mathbb{E}_{\mathcal{D}, q^{\text{rgb}}_{\phi}}&\left[\text{MSE}(\hat{o}_{t+1}, o_{t+1})\right]\\ &+ \beta \mathbb{E}_{\mathcal{D}}\left[D_{\text{KL}}\left(q^{\text{rgb}}_{\phi,\text{kin}} \,\|\, \mathcal{N}(0, I)\right) + D_{\text{KL}}\left(q^{\text{rgb}}_{\phi,\text{res}} \,\|\, \mathcal{N}(0, I)\right)\right].\end{aligned} \tag{6}$$

Here $\mathcal{D}$ denotes the paired training data and β controls the KL regularization strength. The component posteriors $q^{\mathrm{rgb}}_{\phi,\mathrm{kin}}$ and $q^{\mathrm{rgb}}_{\phi,\mathrm{res}}$ are the marginals of $q^{\mathrm{rgb}}_{\phi}(\cdot \mid o_t, o_{t+1})$.

With the skeleton branch frozen, we optimize the RGB encoder and decoder using:

$$\begin{aligned}\mathcal{L}_{\mathrm{latent}} = \mathcal{L}_{\mathrm{rgb}} + \lambda_{\mathrm{ground}}\mathcal{L}_{\mathrm{ground}} + \lambda_{\mathrm{align}}\mathcal{L}_{\mathrm{align}} \\ + \lambda_{\mathrm{swap}}\mathcal{L}_{\mathrm{swap}} + \lambda_{\mathrm{var}}\mathcal{L}_{\mathrm{var}}.\end{aligned} \tag{7}$$

The regularizer $\mathcal{L}_{\mathrm{var}}$ penalizes low per-dimension standard deviations and cross-covariance between the RGB latent components. Grounding and latent alignment are introduced after conditional reconstruction initialization, followed by swap reconstruction. Initialization losses, regularizer definitions, loss weights, and the optimization schedule are provided in Appendix B.1.

3.2 Action-Conditioned World Modeling

Structured world-model conditioning. To preserve a kinematic conditioning pathway for robot action adaptation, we introduce separate projections for the kinematic and residual components. Given a training clip $o_{t:t+H}$, the frozen RGB encoder extracts $z^{\mathrm{rgb}}_{\mathrm{kin},i}$ and $z^{\mathrm{rgb}}_{\mathrm{res},i}$ from each transition (o_i, o_{i+1}), where $i \in \{t, \ldots, t+H-1\}$. We use separate MLP projections P_b with branch-presence masks $m_b \in \{0,1\}$, leaving latent standardization implicit in the notation (Appendix A):

$$\begin{aligned}\tilde{z}_{b,i} &= m_b z^{\mathrm{rgb}}_{b,i}, \qquad b \in \{\mathrm{kin}, \mathrm{res}\}, \\ h_i &= P_{\mathrm{kin}}(\tilde{z}_{\mathrm{kin},i}; m_{\mathrm{kin}}) + P_{\mathrm{res}}(\tilde{z}_{\mathrm{res},i}; m_{\mathrm{res}}),\end{aligned} \tag{8}$$

where a dropped branch has a zero latent and a zero presence mask. The world model $p_\theta(o_{t+1:t+H} \mid o_t, h_{t:t+H-1})$ learns from video-derived latents through full-backbone fine-tuning after projector warmup. Robot-action latents reuse the kinematic projection.

Branch dropout and world-model training. Residual latents use future observations unavailable at deployment. To retain complementary information while preparing for missing conditions, $\mathcal{R}$ samples one pair of branch masks per clip:

$$(m_{\mathrm{kin}}, m_{\mathrm{res}}) = \begin{cases}(1,1), & \text{full (keep)}, \\ (0,1) \text{ or } (1,0), & \text{drop_kin or drop_res (partial)}, \\ (0,0), & \text{drop_both (null)}.\end{cases} \tag{9}$$

These four whole-branch modes share masks across all H transitions; their step-dependent probabilities are given in Appendix B.2. Partial means dropping one entire branch, not individual coordinates; null drops both branches using fixed zeros in standardized latent space. The supervision target remains the complete future clip $o_{t+1:t+H}$ in every mode. The drop_res mode trains prediction from o_t and kinematic latents, matching action-conditioned inference.

Let ρ collect the learnable conditioning parameters. The world-model generation objective is

$$\mathcal{L}_{\mathrm{wm}}(\theta, \rho) = \mathbb{E}_{\mathcal{D}_{\mathrm{video}}, q^{\mathrm{rgb}}_{\phi}, \mathcal{R}, \xi}[\ell_{\mathrm{gen}}(\theta; o_{t+1:t+H}, o_t, h_{t:t+H-1}, \xi)], \tag{10}$$

where $\mathcal{D}_{\mathrm{video}}$ contains training clips, ℓ_{gen} is the generation loss, and ξ collects its stochastic variables. The RGB encoder remains frozen.

Robot action adaptation and inference. To condition the world model on robot controls, we train a lightweight MLP adapter $A_\psi : \mathcal{A} \to \mathbb{R}^{d_{\mathrm{kin}}}$, where $\mathcal{A}$ denotes the robot action space. For each transition (o_i, a_i, o_{i+1}), the frozen RGB encoder provides the kinematic posterior mean $\mu^{\mathrm{rgb}}_{\mathrm{kin},i}$ as the regression target:

$$\mathcal{L}_{\mathrm{adapt}}(\psi) = \mathbb{E}_{\mathcal{D}_{\mathrm{robot}}}\left[d_{\mathrm{adapt}}\left(A_\psi(a_i), \mathrm{sg}\left[\mu^{\mathrm{rgb}}_{\mathrm{kin},i}\right]\right)\right], \tag{11}$$

where $\mathcal{D}_{\mathrm{robot}}$ denotes the robot training data and d_{adapt} is the latent regression criterion. Restricting regression to the kinematic component focuses adaptation on the motion representation learned through skeleton supervision. Only ψ is optimized, with the RGB encoder and (θ, ρ) held fixed.

We then fine-tune the full world-model backbone using predicted kinematic latents and a curriculum residual–mask pair $(\tilde{z}^{(s)}_{\mathrm{res},i}, m^{(s)}_{\mathrm{res}})$ (Appendix B.2):

$$\hat{h}^{\mathrm{act}}_i = P_{\mathrm{kin}}(A_\psi(a_i); 1) + P_{\mathrm{res}}\left(\tilde{z}^{(s)}_{\mathrm{res},i}; m^{(s)}_{\mathrm{res}}\right). \tag{12}$$

The action-conditioned generation objective is

$$\mathcal{L}_{\mathrm{wm}} = \mathbb{E}_{\mathcal{D}_{\mathrm{robot}},\xi}\left[\ell_{\mathrm{gen}}\Big(\theta; o_{t+1:t+H}, o_t, \hat{h}^{\mathrm{act}}_{t:t+H-1}, \xi\Big)\right]. \tag{13}$$

At inference, $\tilde{z}_{\mathrm{res},i} = 0$ and $m_{\mathrm{res}} = 0$, with no future teacher residual; the current observation and actions specify the rollout:

$$\hat{o}_{t+1:t+H} \sim p_\theta\Big(\cdot \mid o_t, \hat{h}^{\mathrm{act}}_{t:t+H-1}\Big). \tag{14}$$

4 EXPERIMENT

We evaluate the effect of the kinematic–residual decomposition on action-conditioned world modeling, and examine motion transfer and rollout evaluation as downstream applications. Specifically, we investigate the following questions:

- **RQ1 (World modeling):** Does adapting robot actions to a dedicated kinematic component improve motion accuracy while preserving visual fidelity? (Section 4.2)
- **RQ2 (Motion representation):** Do skeleton-grounded kinematic latents support more accurate linear decoding of robot motion than unstructured latent actions? (Section 4.3)
- **RQ3 (Motion transfer):** Can a kinematic representation learned from robot interactions transfer human motion to generated robot rollouts in a zero-shot setting? (Section 4.4)
- **RQ4 (Policy evaluation):** How closely do open-loop world-model rollouts reproduce policy rankings and success rates measured in real robot executions? (Section 4.5)

4.1 EXPERIMENTAL SETUP

Data. DROID (Khazatsky et al., 2024) pairs visual observations with robot actions, allowing us to assess whether predicted dynamics follow robot controls. We train the latent action model and world model on DROID, incorporating AgiBot (Bu et al., 2025) to broaden the range of robot interactions observed during training. Evaluation uses 400 DROID clips, each containing one conditioning frame and 12 future frames at 480×640 resolution. More details are provided in Appendix A.

Evaluation Metrics. To quantify trajectory, direction, and magnitude agreement, we report 2D end-effector average displacement error (ADE, px), direction macro-F1 ($\mathrm{F1}_{\mathrm{dir}}$), normalized magnitude MAE (L_1^{mag}), and magnitude Pearson correlation (r_{mag}). PSNR, SSIM, and LPIPS (Zhang et al., 2018) assess pixel, structural, and perceptual fidelity, respectively, with tLPIPS reported as a temporal diagnostic. More details are provided in Appendix C.1.

Implementation Details. We implement the latent action model using Transformer VAEs with $d_{\mathrm{kin}} = d_{\mathrm{res}} = 32$ and use Cosmos-Predict2.5-2B as the world-model backbone. We train with AdamW, using stage-specific learning rates and global batch sizes. SkeLAM training uses NVIDIA A100 GPUs and requires approximately 2 days for one complete run. Architectural specifications, loss weights, training schedules, and inference settings are provided in Appendix B.

Baselines. We compare SkeLAM with latent-action world models DreamDojo (Gao et al., 2026) and CD-LAM (Wei et al., 2026). Alternative conditioning strategies include direct robot actions in IRASim (Zhu et al., 2025), Ctrl-World (Guo et al., 2026), Cosmos-Predict2.5 (Ali et al., 2025), Cosmos3 (Agarwal et al., 2026), and GE-Sim (Liao et al., 2025), and geometric guidance in EnerVerse-AC (Jiang et al., 2025) and OSCAR (Wu & Gao, 2026). TesserAct (Zhen et al., 2025), and GE-Base (Liao et al., 2025) serve as text-conditioned references for visual fidelity, with evaluation details provided in Appendix C.

4.2 ACTION-CONDITIONED WORLD MODELING

Quantitative results. To assess how accurately SkeLAM simulates the consequences of robot actions, we evaluate motion accuracy and visual fidelity on DROID. Table 1 shows that SkeLAM ranks first on seven of eight metrics among the reported action-conditioned models, including all four visual fidelity metrics. Compared with DreamDojo and CD-LAM, SkeLAM reduces ADE by 19.5% and 23.7%, respectively, alongside LPIPS reductions of 11.1% and 17.4%. These improvements

indicate closer agreement with both the target robot trajectories and visual observations. Ctrl-World and OSCAR achieve higher direction F1, while SkeLAM attains lower trajectory error and stronger visual fidelity. In particular, compared with Ctrl-World, SkeLAM reduces ADE from 42.96 to 41.15 px and LPIPS from 0.4029 to 0.1310. The combined improvements in trajectory accuracy and visual fidelity support the generation of robot motion together with the associated scene dynamics.

Table 1: **World modeling results.** Control denotes the inference-time conditioning interface. Text-conditioned models serve as visual references. **Bold** and underlined values indicate the best and second-best action-conditioned results. † denotes large-scale interaction pretraining.

		Visual fidelity				Motion accuracy			
Method	Control	PSNR↑	SSIM↑	LPIPS↓	MSE↓	ADE↓	$\mathrm{F1}_{\mathrm{dir}}$ ↑	L_1^{mag} ↓	r_{mag} ↑
Text-conditioned references									
TesserAct	Text	15.08	0.7400	0.2561	0.0349	83.76	0.2950	0.0797	0.0823
GE-Base	Text	12.51	0.4530	0.5864	0.0596	55.30	0.3006	0.0956	0.0504
Action- and geometry-conditioned models									
IRASim	Action emb.	11.55	0.4137	0.5993	0.0738	67.11	0.2616	0.0843	0.1193
Ctrl-World	Action emb.	16.34	0.5910	0.4029	0.0255	42.96	**0.5888**	0.0511	0.5154
Cosmos-Predict2.5	Action emb.	11.51	0.4209	0.6018	0.0758	60.76	0.2375	0.0839	0.1049
Cosmos3	Action emb.	16.98	0.7630	0.2186	0.0270	67.23	0.3278	0.0725	0.2518
EnerVerse-AC	Action map	14.13	0.5157	0.4139	0.0415	59.63	0.2433	0.0982	-0.0050
GE-Sim	Action map	12.53	0.4550	0.4950	0.0593	55.57	0.2799	0.0941	0.0091
OSCAR	Skeleton	12.57	0.4471	0.4788	0.0587	43.74	0.5810	0.0530	0.5029
Models with learned latent actions									
DreamDojo†	a	18.91	0.8057	0.1473	0.0168	51.11	0.4989	0.0667	0.3335
CD-LAM†	$a \to z$	18.34	0.7910	0.1586	0.0181	53.90	0.4903	0.0690	0.3179
SkeLAM (ours)	$a \to z_{\mathrm{kin}}$	**19.35**	**0.8340**	**0.1310**	**0.0151**	**41.15**	0.5465	**0.0510**	**0.5277**

Qualitative results. To evaluate whether generated rollouts reproduce robot motion while preserving scene structure, we compare predictions with ground truth in Figure 3. The overlaid trajectories reveal deviations from the reference motion in DreamDojo and OSCAR, accompanied by visible arm deformation in later frames. Ctrl-World produces a shorter trajectory and local gripper distortions, while Cosmos-Predict2.5 alters the apparent table scale and bowl position. SkeLAM more closely follows the curved reference trajectory while retaining the tabletop layout, although local gripper errors remain. This example highlights the importance of assessing motion correspondence alongside scene consistency. Two additional comparisons appear in Appendix E.

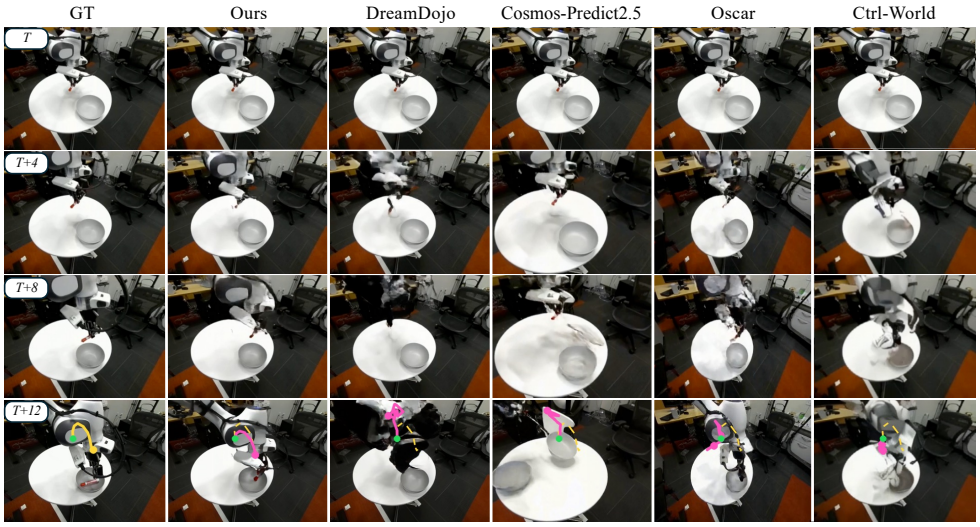


Figure 3: **Qualitative comparison of action-conditioned world modeling.** Ground-truth and generated observations are compared at selected time steps, highlighting differences in end-effector motion, manipulated-object geometry, and scene appearance.

Table 2: **Component ablations.** Each variant removes one component from the full model under the same training budget. **Bold** and underlined values indicate the best and second-best results.

	Visual fidelity				Motion accuracy			
Variant	PSNR↑	SSIM↑	LPIPS↓	MSE↓	ADE↓	$\mathrm{F1}_{\mathrm{dir}}$ ↑	L_1^{mag} ↓	r_{mag} ↑
Full SkeLAM	19.35	0.8340	0.1310	0.0151	**41.15**	**0.5465**	**0.0510**	**0.5277**
w/o skeleton supervision	18.18	0.8206	0.1538	0.0188	64.59	0.3268	0.0723	0.2009
w/o cross-modal swap	19.30	0.8330	0.1436	0.0192	66.74	0.3132	0.0752	0.1897
w/o residual latent	**19.38**	**0.8348**	**0.1305**	**0.0150**	43.37	0.5263	0.0547	0.4737
w/o action adaptation	18.14	0.8196	0.1545	0.0189	65.84	0.3259	0.0739	0.1603

Ablation studies. To assess how each component supports action-conditioned generation, we compare variants in Table 2. **(i) Skeleton supervision** grounds the RGB kinematic latent in robot motion; removing it increases ADE from 41.15 to 64.59 and reduces magnitude correlation from 0.5277 to 0.2009, supporting geometric supervision for learning motion representations suitable for action conditioning. **(ii) Cross-modal swap** couples skeleton supervision to RGB generation beyond latent alignment; removing it increases ADE to 66.74 and reduces magnitude correlation to 0.1897, while LPIPS rises from 0.1310 to 0.1436, supporting reconstruction-level RGB–skeleton alignment for downstream action conditioning. **(iii) Residual latent** retains complementary scene dynamics; zeroing it from world-model pretraining maintains similar visual fidelity (0.1305 vs. 0.1310 LPIPS) while ADE increases to 43.37 and magnitude correlation decreases to 0.4737, suggesting that complementary visual information benefits motion conditioning. **(iv) Action adaptation** supervises the action-to-kinematic mapping; omitting adapter regression increases ADE to 65.84, supporting explicit supervision of the mapping from robot actions to the learned kinematic representation. Further details and adaptation-budget analysis are provided in Appendix D.

4.3 Motion Representation Analysis

Motion decodability. To examine how robot motion is encoded, we train linear probes on frozen latent representations. As shown in Table 3, the kinematic component achieves an EEF MAE of 5.99×10^{-3} and a magnitude macro-F1 of 0.7467, improving on both latent-action baselines. Within SkeLAM, it also outperforms the residual component on all three metrics. Concatenating the residual increases EEF R^2 from 0.3200 to 0.3277 and magnitude macro-F1 from 0.7467 to 0.7482. The kinematic component thus approaches the probing performance of the full representation using half the latent dimensions, supporting its specialization for displacement and magnitude encoding.

Cross-modal compatibility. To assess cross-modal motion conditioning, we intervene on the latent inputs to the RGB decoder. Figure 4 shows that substituting paired and mismatched skeleton kinematic means increases reconstruction MAE by 0.53% and 11.67%, respectively. SSIM exhibits the same ordering. Zeroing the kinematic component increases MAE by 5.78%, compared with 0.61% for the residual. The greater degradation under mismatched substitution than kinematic removal indicates sensitivity to motion correspondence. These results support compatible RGB and skeleton conditioning and a functional role for the kinematic component in reconstruction.

4.4 Zero-Shot Human-to-Robot Motion Transfer

To examine whether the learned motion representation transfers across embodiments, we use human demonstration videos from UniSkill (Kim et al., 2025) as motion prompts. The RGB encoder extracts latent actions from these videos to directly condition SkeLAM, with a robot observation specifying the initial scene. All model parameters remain fixed during transfer. We compare with the pretrained checkpoint of DreamDojo (Gao et al., 2026), which supports latent-action conditioning. As shown in Figure 5, SkeLAM generates distinct end-effector motions corresponding to the human demonstrations: a downward reach for the towel-pushing prompt and an upward movement for the tissue-pulling prompt. DreamDojo exhibits limited end-effector displacement in both examples. These results provide qualitative evidence that human-derived latent actions can guide robot motion despite differences in embodiment and scene appearance.

Table 3: **Motion probing.** Linear probes predict EEF displacement (MAE, R^2) and motion magnitude (macro-F1) from frozen representations. Latents are 32D except the 64D concatenation. †: privileged skeleton input.

Representation	MAE↓	R^2_{EEF}↑	F1_{mag}↑
DreamDojo	0.01620	0.0078	0.7183
CD-LAM	0.01615	0.0079	0.7006
SkeLAM z_{kin}	0.00599	0.3200	0.7467
SkeLAM z_{res}	0.00678	0.1879	0.5652
$[z_{\text{kin}}; z_{\text{res}}]$	0.00597	0.3277	0.7482
Skeleton teacher†	0.00552	0.4312	0.7233

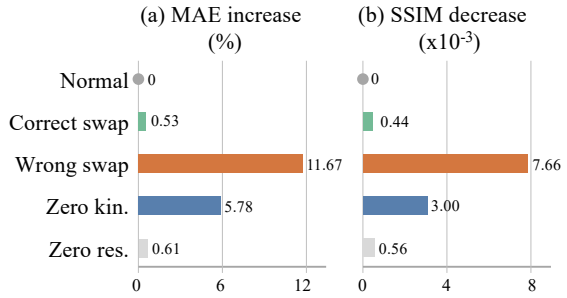


Figure 4: **Latent action interventions.** Reconstruction MAE increase and SSIM decrease relative to Normal. Correct and wrong swaps use paired and mismatched skeleton kinematic means.

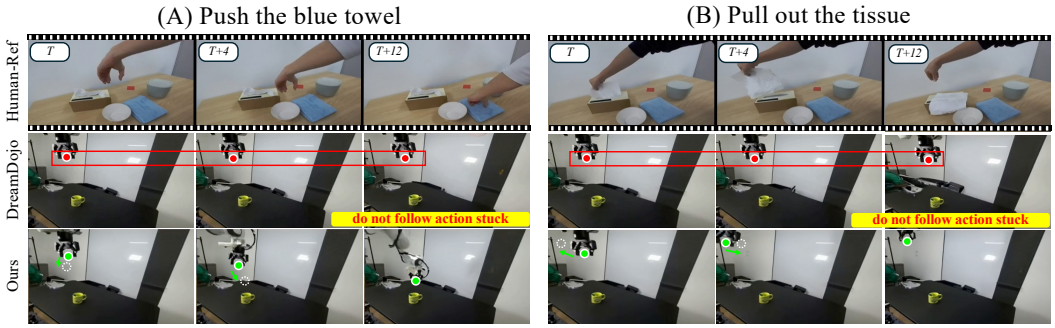


Figure 5: **Zero-shot human-to-robot motion transfer.** Human videos from UniSkill (Kim et al., 2025) provide latent conditions for robot rollout generation. Markers indicate end-effector positions.

Table 4: **Open-loop policy evaluation on RoboArena.** Seven policies are evaluated across 15 sessions (105 episodes). Episode-level metrics include balanced accuracy (BA, %), success recall (%), and Matthews correlation coefficient (MCC). Policy-level metrics include Spearman rank correlation (ρ), Pearson correlation (r), and success-rate MAE (Δ_{SR}).

	Episode-level agreement			Policy-level agreement		
Model	BA↑	Recall↑	MCC↑	ρ ↑	r ↑	Δ_{SR} ↓
Cosmos-Predict2.5	48.44	0.00	-0.0525	-0.0756	-0.1936	7.62
SkeLAM	63.02	33.33	0.2483	0.5347	0.6091	4.76

4.5 World Model for Robot Policy Evaluation

To assess whether generated rollouts preserve task outcomes and relative policy performance, we evaluate seven policies across 15 RoboArena sessions (105 episodes). Both models receive the same initial observations and recorded actions. Human annotators label task success in generated rollouts for comparison with real execution outcomes. Table 4 reports 33.33% success recall and 63.02% balanced accuracy for SkeLAM, compared with 0.00% and 48.44% for Cosmos-Predict2.5. At the policy level, success-rate MAE decreases from 7.62 to 4.76 percentage points, with positive Spearman (ρ=0.5347) and Pearson (r=0.6091) correlations for SkeLAM. Together, these metrics indicate improved agreement in both absolute policy success rates and relative policy ordering. Figure 6 compares predicted and real success rates for individual policies, with proximity to the diagonal indicating closer agreement. The results support agreement in both individual task outcomes and relative policy performance under open-loop action replay. Nevertheless, the modest success recall indicates that reproducing successful task outcomes remains challenging. Further evaluation details are provided in Appendix C.3.

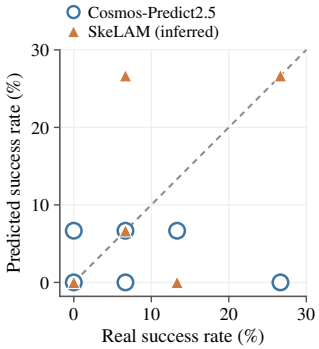


Figure 6: **Policy-level agreement.** The dashed line indicates perfect agreement.

5 Conclusion

We presented SkeLAM, a skeleton-guided latent action model for controllable robot world modeling. The kinematic component provides a motion-grounded target for robot action adaptation, while the residual retains complementary visual dynamics. Cross-modal swap reconstruction and branch dropout support reuse of the learned kinematic pathway for action-conditioned generation. Experiments on DROID show improved motion accuracy and visual fidelity, with representation analyses supporting kinematic specialization. Qualitative results demonstrate zero-shot human-to-robot motion transfer, while policy evaluation yields success-rate estimates consistent with real robot executions. These results support structuring latent actions around robot motion while retaining complementary information for generation. Future work will extend the current recorded-action replay protocol to closed-loop policy evaluation. A key question is how prediction errors affect subsequent policy decisions and task outcomes over longer interaction horizons.

References

Niket Agarwal, Arslan Ali, Jon Allen, Martin Antolini, Adeline Aubame, Alisson Azzolini, Junjie Bai, Maciej Bala, Yogesh Balaji, Josh Bapst, et al. Cosmos 3: Omnimodal world models for physical ai. *arXiv preprint arXiv:2606.02800*, 2026.

Arslan Ali, Junjie Bai, Maciej Bala, Yogesh Balaji, Aaron Blakeman, Tiffany Cai, Jiaxin Cao, Tianshi Cao, Elizabeth Cha, Yu-Wei Chao, et al. World simulation with video foundation models for physical ai. *arXiv preprint arXiv:2511.00062*, 2025.

Jake Bruce, Michael D Dennis, Ashley Edwards, Jack Parker-Holder, Yuge Shi, Edward Hughes, Matthew Lai, Aditi Mavalankar, Richie Steigerwald, Chris Apps, Yusuf Aytar, Sarah Maria Elisabeth Bechtle, Feryal Behbahani, Stephanie C.Y. Chan, Nicolas Heess, Lucy Gonzalez, Simon Osindero, Sherjil Ozair, Scott Reed, Jingwei Zhang, Konrad Zolna, Jeff Clune, Nando De Freitas, Satinder Singh, and Tim Rocktäschel. Genie: Generative interactive environments. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 4603–4623. PMLR, 2024. URL https://proceedings.mlr.press/v235/bruce24a.html.

Qingwen Bu, Jisong Cai, Li Chen, Xiuqi Cui, Yan Ding, Siyuan Feng, Xindong He, Xu Huang, et al. AgiBot World Colosseo: A large-scale manipulation platform for scalable and intelligent embodied systems. In *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2025. URL https://arxiv.org/abs/2503.06669.

Xizhou Bu, Qingda Hu, Lei Zhou, Lingfeng Zhang, Yingbo Tang, Zihao Liu, Xinyi Tao, Zhiqiang Ma, Qingqiu Huang, Chufeng Tang, Hongbo Wang, Jing Zhang, Jiayi Ma, Hangjun Ye, Wei Li, and Xiaoshuai Hao. What Matters for Latent Actions in Robot Learning. *arXiv preprint arXiv:2608.19613*, 2026a. URL https://arxiv.org/abs/2608.19613.

Xizhou Bu, Jiexi Lyu, Fulei Sun, Ruichen Yang, Zhiqiang Ma, and Wei Li. LAOF: Robust Latent Action Learning with Optical Flow Constraints. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 27334–27344, 2026b. URL https://arxiv.org/abs/2511.16407.

Junhao Cai, Zetao Cai, Jiafei Cao, Yilun Chen, Zeyu He, Lei Jiang, Hang Li, Hengjie Li, Yang Li, Yufei Liu, Yanan Lu, Qi Lv, Haoxiang Ma, Jiangmiao Pang, Yu Qiao, Zherui Qiu, Yanqing Shen, Xu Shi, Yang Tian, Bolun Wang, Hanqing Wang, Jiaheng Wang, Tai Wang, Xueyuan Wei, Chao Wu, Yiman Xie, Boyang Xing, Yuqiang Yang, Yuyin Yang, Qiaojun Yu, Feng Yuan, Jia Zeng, Jingjing Zhang, Shenghan Zhang, Shi Zhang, Zhuoma Zhaxi, Bowen Zhou, Yuanzhen Zhou, Yunsong Zhou, Hongrui Zhu, Yangkun Zhu, and Yuchen Zhu. InternVLA-A1: Unifying Understanding, Generation and Action for Robotic Manipulation. *arXiv preprint arXiv:2601.02456*, 2026a. URL https://arxiv.org/abs/2601.02456.

Zetao Cai, Yaping Li, Yiqun Wang, Xinyu Zhan, Yuyin Yang, Haoxiang Ma, Kailin Li, Tao Lu, Jiangmiao Pang, Linning Xu, and Dahua Lin. Skel-WAM: A Hand-Skeleton-Conditioned World Action Model for Human-to-Robot Manipulation Transfer. *arXiv preprint arXiv:2609.21514*, 2026b. URL https://arxiv.org/abs/2609.21514.

Zhijing Cheng, Xuancheng Zhang, Donglin Di, Lei Fan, Baorui Ma, Hao Li, and Xun Yang. MOJITO: Modal joint learning for unified end-to-end autonomous driving. In *European Conference on Computer Vision*, pp. 620–639. Springer, 2026a.

Zhijing Cheng, Xuancheng Zhang, Donglin Di, Chen Wei, Hao Li, and Xun Yang. MoCa: Modeling object consistency for 3D camera control in video generation. In *The Fourteenth International Conference on Learning Representations*, 2026b.

Shenyuan Gao, Siyuan Zhou, Yilun Du, Jun Zhang, and Chuang Gan. AdaWorld: Learning adaptable world models with latent actions. In *International Conference on Machine Learning*, 2025. URL https://arxiv.org/abs/2503.18938.

Shenyuan Gao, William Liang, Kaiyuan Zheng, Ayaan Malik, Seonghyeon Ye, Sihyun Yu, Wei-Cheng Tseng, Yuzhu Dong, Kaichun Mo, Chen-Hsuan Lin, Qianli Ma, Seungjun Nah, Loic Magne, Jiannan Xiang, Yuqi Xie, Ruijie Zheng, Dantong Niu, You Liang Tan, K. R. Zentner, George Kurian, Suneel Indupuru, Pooya Jannaty, Jinwei Gu, Jun Zhang, Jitendra Malik, Pieter Abbeel, Ming-Yu Liu, Yuke Zhu, Joel Jang, and Linxi "Jim" Fan. DreamDojo: A generalist robot world model from large-scale human videos. *arXiv preprint arXiv:2602.06949*, 2026. URL https://arxiv.org/abs/2602.06949.

Quentin Garrido, Tushar Nagarajan, Basile Terver, Nicolas Ballas, Yann LeCun, and Michael Rabbat. Learning Latent Action World Models In The Wild. *arXiv preprint arXiv:2601.05230*, 2026. URL https://arxiv.org/abs/2601.05230.

Manish Kumar Govind, Dominick Reilly, Pu Wang, and Srijan Das. UniLACT: Depth-Aware RGB Latent Action Learning for Vision-Language-Action Models. *arXiv preprint arXiv:2602.20231*, 2026. URL https://arxiv.org/abs/2602.20231.

Chenghao Gu, Hanyang Yu, Jingbo Zhang, Haitao Lin, Wenyao Zhang, Jinghe Wang, Hanglei Jin, Shuzhao Xie, Jingyan Jiang, and Zhi Wang. GeniWorld: A Generalizable Interactive World Model for Robotic Manipulation via Visual Actions. *arXiv preprint arXiv:2608.06332*, 2026. URL https://arxiv.org/abs/2608.06332.

Yanjiang Guo, Lucy Xiaoyang Shi, Jianyu Chen, and Chelsea Finn. Ctrl-World: A Controllable Generative World Model for Robot Manipulation. In *International Conference on Learning Representations*, 2026. URL https://arxiv.org/abs/2510.10125.

Yan Huang, Meimei Zhang, Hong Lei, Chenyang Ji, Jiayi Wang, and Huidong Zhu. CE-Agent: a cloud-edge collaborative agent framework with dynamic offloading and synergistic distillation. *Journal of Cloud Computing*, 2026. doi: 10.1186/s13677-026-00984-5. URL https://doi.org/10.1186/s13677-026-00984-5.

Yuxin Jiang, Shengcong Chen, Siyuan Huang, Liliang Chen, Pengfei Zhou, Yue Liao, Xindong He, Chiming Liu, Hongsheng Li, Maoqing Yao, and Guanghui Ren. Enerverse-ac: Envisioning embodied environments with action condition. *arXiv preprint arXiv:2505.09723*, 2025.

Yuxin Jiang, Yuchao Gu, Ivor W. Tsang, and Mike Zheng Shou. Olaf-World: Orienting Latent Actions for Video World Modeling. In *International Conference on Machine Learning*, 2026. URL https://arxiv.org/abs/2602.10104.

Alexander Khazatsky, Karl Pertsch, Suraj Nair, Ashwin Balakrishna, Sudeep Dasari, Siddharth Karamcheti, Soroush Nasiriany, Mohan Kumar Srirama, Lawrence Yunliang Chen, Kirsty Ellis, Peter David Fagan, Joey Hejna, Masha Itkina, Marion Lepert, Yecheng Jason Ma, Patrick Tree Miller, Jimmy Wu, Suneel Belkhale, Shivin Dass, Huy Ha, Arhan Jain, Abraham Lee, Youngwoon Lee, Marius Memmel, Sungjae Park, Ilija Radosavovic, Kaiyuan Wang, Albert Zhan, Kevin Black, Cheng Chi, Kyle Beltran Hatch, Shan Lin, Jingpei Lu, Jean Mercat, Abdul Rehman, Pannag R Sanketi, Archit Sharma, Cody Simpson, Quan Vuong, Homer Rich Walke, Blake Wulfe, Ted Xiao, Jonathan Heewon Yang, Arefeh Yavary, Tony Z. Zhao, Christopher Agia, Rohan Baijal, Mateo Guaman Castro, Daphne Chen, Qiuyu Chen, Trinity Chung, Jaimyn Drake, Ethan Paul Foster, Jensen Gao, Vitor Guizilini, David Antonio Herrera, Minho Heo, Kyle Hsu, Jiaheng Hu, Muhammad Zubair Irshad, Donovon Jackson, Charlotte Le, Yunshuang Li, Kevin Lin, Roy Lin, Zehan Ma, Abhiram Maddukuri, Suvir Mirchandani, Daniel Morton, Tony Nguyen, Abigail

O'Neill, Rosario Scalise, Derick Seale, Victor Son, Stephen Tian, Emi Tran, Andrew E. Wang, Yilin Wu, Annie Xie, Jingyun Yang, Patrick Yin, Yunchu Zhang, Osbert Bastani, Glen Berseth, Jeannette Bohg, Ken Goldberg, Abhinav Gupta, Abhishek Gupta, Dinesh Jayaraman, Joseph J Lim, Jitendra Malik, Roberto Martín-Martín, Subramanian Ramamoorthy, Dorsa Sadigh, Shuran Song, Jiajun Wu, Michael C. Yip, Yuke Zhu, Thomas Kollar, Sergey Levine, and Chelsea Finn. DROID: A large-scale in-the-wild robot manipulation dataset. *arXiv preprint arXiv:2403.12945*, 2024. URL https://arxiv.org/abs/2403.12945.

Hanjung Kim, Jaehyun Kang, Hyolim Kang, Meedeum Cho, Seon Joo Kim, and Youngwoon Lee. Uniskill: Imitating human videos via cross-embodiment skill representations. In *9th Annual Conference on Robot Learning*, 2025.

Hao Li, Qi Lv, Rui Shao, Xiang Deng, Yinchuan Li, Jianye Hao, and Liqiang Nie. STAR: Learning Diverse Robot Skill Abstractions through Rotation-Augmented Vector Quantization. In *Proceedings of the 42nd International Conference on Machine Learning*, pp. 35562–35580, 2025. URL https://proceedings.mlr.press/v267/li25bo.html.

Anthony Liang, Pavel Czempin, Matthew M. Hong, Yutai Zhou, Jingzhen Wang, Erdem Biyik, and Stephen Tu. CLAM: Continuous latent action models for robot learning from unlabeled demonstrations. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2026. URL https://arxiv.org/abs/2505.04999.

Yue Liao, Pengfei Zhou, Siyuan Huang, Donglin Yang, Shengcong Chen, Yuxin Jiang, Yue Hu, Jingbin Cai, Si Liu, Jianlan Luo, Liliang Chen, Shuicheng Yan, Maoqing Yao, and Guanghui Ren. Genie Envisioner: A Unified World Foundation Platform for Robotic Manipulation. *arXiv preprint arXiv:2508.05635*, 2025. URL https://arxiv.org/abs/2508.05635.

Jiahao Liu, Zhongpu Xia, Shuai Tian, Huangrui Li, Yuhang Zheng, Ning Ma, Xin Fu, Xiaotian Liu, Jing Li, Yixian Li, ShangQing Zhou, Zebin Xing, Linbo Wang, Chaoyue Li, Haoran Li, and Dongbin Zhao. WALA Learning Executable Latent Actions from Action-Labeled Demonstrations and Action-Free Videos. *arXiv preprint arXiv:2607.11397*, 2026a. URL https://arxiv.org/abs/2607.11397.

Yichen Liu, Puzhen Yuan, Xiang Zhu, Yanjiang Guo, and Jianyu Chen. GALA: Geometry-Aware Latent Action Modeling for Vision-Language-Action Model Pretraining across Embodiments. *arXiv preprint arXiv:2609.21948*, 2026b. URL https://arxiv.org/abs/2609.21948.

Lianghao Luo, Xizhou Bu, Ruyan Liu, Qingqiu Huang, Chufeng Tang, Xiaoshuai Hao, Hongbo Wang, and Wei Li. Data-Asymmetric Latent Imagination and Reranking for 3D Robotic Imitation Learning. *arXiv preprint arXiv:2605.10166*, 2026. URL https://arxiv.org/abs/2605.10166.

Qi Lv, Weijie Kong, Hao Li, Jia Zeng, Zherui Qiu, Delin Qu, Haoming Song, Qizhi Chen, Xiang Deng, and Jiangmiao Pang. F1: A Vision-Language-Action Model Bridging Understanding and Generation to Actions. *arXiv preprint arXiv:2509.06951*, 2025a. URL https://arxiv.org/abs/2509.06951.

Qi Lv, Hao Li, Xiang Deng, Rui Shao, Yinchuan Li, Jianye Hao, Longxiang Gao, Michael Yu Wang, and Liqiang Nie. Spatial-Temporal Graph Diffusion Policy with Kinematic Modeling for Bimanual Robotic Manipulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 17394–17404, 2025b. URL https://arxiv.org/abs/2503.10743.

Qi Lv, Jianming Xing, Zhao Yang, Mingyuan Yao, Yinan Shi, Yawei Jueluo, Mike Zheng Shou, and Xiang Deng. Hermite Curves as Trajectory Priors for Vision-Language-Action Models. *arXiv preprint arXiv:2608.01265*, 2026. URL https://arxiv.org/abs/2608.01265.

Jiexi Lyu, Xizhou Bu, Qingqiu Huang, Chufeng Tang, Xiaoshuai Hao, Hongbo Wang, and Wei Li. LAFP: Preserving Latent Action Structure in Latent Policy Learning via Flow Matching. *arXiv preprint arXiv:2606.10517*, 2026. URL https://arxiv.org/abs/2606.10517.

Dujun Nie, Fengjiao Chen, Qi Lv, Jun Kuang, Xiaoyu Li, Xuezhi Cao, and Xunliang Cai. LARY: A Latent Action Representation Yielding Benchmark for Generalizable Vision-to-Action Alignment. *arXiv preprint arXiv:2604.11689*, 2026. URL https://arxiv.org/abs/2604.11689.

Alexander Nikulin, Ilya Zisman, Denis Tarasov, Nikita Lyubaykin, Andrei Polubarov, Igor Kiselev, and Vladislav Kurenkov. Latent action learning requires supervision in the presence of distractors. In *International Conference on Machine Learning*, 2025. URL https://arxiv.org/abs/2502.00379.

Zhenhao Shen, Jiaqi Liang, Jasper Lu, Feng Jiang, Yuran Wang, Chuanbo Wei, Jiayi Liu, Jianchun Yang, Qize Yu, Jiadi You, Ce Hao, Guanqi He, Chen Xie, and Ruihai Wu. LD4WAM: Learning Latent Dynamics from Human Videos for World Action Models. *arXiv preprint arXiv:2608.22403*, 2026. URL https://arxiv.org/abs/2608.22403.

Juyi Sheng, Hua Wang, and Mengyuan Liu. SkeleWAM: Skeleton World-Action Modeling for Efficient Robotic Manipulation. *arXiv preprint arXiv:2610.02120*, 2026. URL https://arxiv.org/abs/2610.02120.

Boyang Wang, Haoran Zhang, Shujie Zhang, Jinkun Hao, Mingda Jia, Qi Lv, Yucheng Mao, Zhaoyang Lyu, Jia Zeng, Xudong Xu, and Jiangmiao Pang. RoboVIP: Multi-View Video Generation with Visual Identity Prompting Augments Robot Manipulation. *arXiv preprint arXiv:2601.05241*, 2026. URL https://arxiv.org/abs/2601.05241.

Lirui Wang, Kevin Zhao, Chaoqi Liu, and Xinlei Chen. Learning Real-World Action-Video Dynamics with Heterogeneous Masked Autoregression. *arXiv preprint arXiv:2502.04296*, 2025a. URL https://arxiv.org/abs/2502.04296.

Yuang Wang, Chao Wen, Haoyu Guo, Sida Peng, Minghan Qin, Hujun Bao, Xiaowei Zhou, and Ruizhen Hu. Precise action-to-video generation through visual action prompts. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025b. URL https://arxiv.org/abs/2508.13104.

Yufan Wei, Kun Zhou, Lingjun Mao, Zijun Zhang, Ziming Xu, Ziqiao Xi, Shuang Liang, Ruobing Han, Yuchen Yan, Xinyue Wang, Fan Feng, and Biwei Huang. Causally debiased latent action model for embodied action conditioned world models. *arXiv preprint arXiv:2607.09185*, 2026.

Jialong Wu, Shaofeng Yin, Ningya Feng, Xu He, Dong Li, Jianye Hao, and Mingsheng Long. iVideoGPT: Interactive VideoGPTs are scalable world models. In *Advances in Neural Information Processing Systems*, 2024. URL https://arxiv.org/abs/2405.15223.

Zhuoyuan Wu and Jun Gao. OSCAR: Omni-Embodiment Action-Conditioned World Model for Robotics. *arXiv preprint arXiv:2606.04463*, 2026. URL https://arxiv.org/abs/2606.04463.

Yifan Xie, Hekun Tian, Jinkun Liu, YuAn Wang, Qiao Sun, and Wenbo Ding. GeoLAM: Learning Geometry-Grounded Latent Actions from Unlabeled Human Videos. *arXiv preprint arXiv:2609.17099*, 2026. URL https://arxiv.org/abs/2609.17099.

Mutian Xu, Tianbao Zhang, Tianqi Liu, Zhaoxi Chen, Xiaoguang Han, and Ziwei Liu. Kinema4D: Kinematic 4D World Modeling for Spatiotemporal Embodied Simulation. *arXiv preprint arXiv:2603.16669*, 2026. URL https://arxiv.org/abs/2603.16669.

Jiange Yang, Yansong Shi, Haoyi Zhu, Mingyu Liu, Kaijing Ma, Yating Wang, Gangshan Wu, Tong He, and Limin Wang. CoMo: Learning Continuous Latent Motion from Internet Videos for Scalable Robot Learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026a. URL https://arxiv.org/abs/2505.17006.

Jiarui Yang, Jiale Zhange, Jiawei Li, Hang Guo, Wen Huang, Jinpeng Wang, Peidong Liu, and Shu-Tao Xia. LAWM-3D: Learning 3D-Aware Latent Actions from Human Videos for Generalizable Robot World Models. *arXiv preprint arXiv:2608.05706*, 2026b. URL https://arxiv.org/abs/2608.05706.

Pei Yang, Hai Ci, Yanzhe Chen, Qi Lv, Han Cai, and Mike Zheng Shou. ActionMap: Robot Policy Learning via Voxel Action Heatmap. *arXiv preprint arXiv:2606.06904*, 2026c. URL https://arxiv.org/abs/2606.06904.

Sherry Yang, Yilun Du, Seyed Ghasemipour, Jonathan Tompson, Leslie Kaelbling, Dale Schuurmans, and Pieter Abbeel. Learning interactive real-world simulators. In *International Conference on Learning Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/c4d66eae503694424123b93ac0fbaf17-Abstract-Conference.html.

Seonghyeon Ye, Joel Jang, Byeongguk Jeon, Se June Joo, Jianwei Yang, Baolin Peng, Ajay Mandlekar, Reuben Tan, Yu-Wei Chao, Bill Yuchen Lin, Lars Liden, Kimin Lee, Jianfeng Gao, Luke Zettlemoyer, Dieter Fox, and Minjoon Seo. Latent action pretraining from videos. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/45d74e190008c7bff2845ffc8e3facd3-Abstract-Conference.html.

Zhentong Ye, Lei Zhang, Sijia Zhou, Yingda Yu, Yuehan Shi, Jiaqi Xuan, Shuaiwu Dong, Guanchao Tong, Meimei Zhang, and Bin Li. OSEF: One-step evidence fusion for cross-video scene procedure planning. *arXiv preprint arXiv:2607.29401*, 2026. doi: 10.48550/arXiv.2607.29401. URL https://arxiv.org/abs/2607.29401.

Chuheng Zhang, Tim Pearce, Pushi Zhang, Kaixin Wang, Xiaoyu Chen, Wei Shen, Li Zhao, and Jiang Bian. What do latent action models actually learn? In *Advances in Neural Information Processing Systems*, 2025. URL https://arxiv.org/abs/2506.15691.

Meimei Zhang, Xuetao Zhang, Hong Lei, Ziyang Xu, Xuan Qi, Sa Liu, and Jiarui Ruan. Difficulty-aware instructional step reasoning for temporal answer grounding in single video. In *NLPCC 2026 Shared Tasks*, 2026. URL https://openreview.net/forum?id=ZX1jk0WsM7. Accepted, to appear.

Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *CVPR*, 2018.

Haoyu Zhen, Qiao Sun, Hongxin Zhang, Junyan Li, Siyuan Zhou, Yilun Du, and Chuang Gan. Tesseract: learning 4d embodied world models. *arXiv preprint arXiv:2504.20995*, 2025.

Fangqi Zhu, Hongtao Wu, Song Guo, Yuxiao Liu, Chilam Cheang, and Tao Kong. IRASim: A fine-grained world model for robot manipulation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025. URL https://arxiv.org/abs/2406.14540.

A Data and Preprocessing

Data and splits. DROID and AgiBot World (Bu et al., 2025) are used in latent-action learning, world-model training, and action adaptation. DROID contributes 60,000 training clips to Stage 1 and 45,358 clips to each of Stages 2A, 2B, 3A, and 3B. RGB and skeleton observations are matched by episode, camera, and frame index; overlapping windows from an episode remain in the same partition. Normalization statistics are fitted on training data only. The DROID evaluation selects one clip from each of 400 trajectories: 100 trajectories per motion quartile, with additional camera stratification. AgiBot evaluation uses 400 held-out clips and 800 arm trajectories.

Preprocessing. RGB and pre-rendered skeleton observations come from the corresponding OSCAR (Wu & Gao, 2026) subsets, following its kinematics, projection, and rasterization protocol. Stage 1 letterboxes both modalities to 240×320 (height $\times$ width): resize by $\min(240/H_o, 320/W_o)$ using OpenCV `INTER_AREA`, round the resized dimensions, and center on a zero canvas, assigning odd padding pixels to the bottom or right. Intensities are divided by 255. The target skeleton mask selects pixels with an 8-bit RGB channel sum above 15, then applies one dilation with a 5×5 elliptical kernel. It is used only in the skeleton loss. Stage 2/3 uses a $4{:}3$ center crop resized to 480×640 for video generation; the frozen latent encoder keeps the Stage 1 resolution and uses the same source frame indices.

Temporal alignment and action interface. A DROID window samples 13 frames from 60-FPS video at indices $f_k = f_0 + 4k$, $k = 0, \ldots, 12$, giving 15 FPS and a 0.8-second span; consecutive window starts differ by 24 source frames. The first frame conditions 12 future frames. Each transition pairs (o_k, o_{k+1}) with an executed state-delta proxy

$$a_k = [p_{k+1} - p_k;\ \Delta\omega_k;\ g_{k+1}] \in \mathbb{R}^7,$$

where translation is in the robot base frame, $\Delta\omega_k$ is the rotation vector of the relative $\mathrm{SO}(3)$ rotation, and g_{k+1} is destination gripper openness. Deltas are between consecutive sampled states, not relative to the clip start or joint-space commands. Manifests retain source indices, timestamps, and transition-validity indicators.

Deterministic latent extraction. The frozen Stage 1 RGB encoder exports posterior means as two 12×32 arrays, z_{kin} and z_{res}; the skeleton encoder is not used for world-model conditioning. Each component is standardized elementwise as $\bar{z} = (z - m_z) \oslash \max(s_z, \epsilon)$ using training statistics. The same kinematic statistics define adapter targets and world-model inputs; normalization is implicit in the main-text notation. Offline caches retain episode, camera, frame, preprocessing, and checkpoint identifiers and receive no gradients.

B Model and Training Details

Architecture and initialization. Both Transformer-VAE branches start from random weights and use 16×16 patches, width 1,024, 16 encoder and 16 decoder blocks, 16 attention heads, causal temporal attention, and zero Transformer dropout. The RGB posterior has 32 kinematic and 32 residual dimensions; the skeleton posterior has 32 kinematic dimensions. The world model starts from official Cosmos-Predict2.5-2B weights (Ali et al., 2025), independently of the DreamDojo baseline. The final Stage 1 teacher, video tokenizer, and text encoder remain frozen throughout Stage 2/3.

Figure 7 summarizes the training stages, selected checkpoints, and dependencies.

B.1 Latent-action training

Stages 1A/B/C initialize RGB and skeleton reconstruction, add grounding and alignment, then add swap, respectively (Figure 7; K = 1,000). The skeleton branch is frozen in Stage 1B/C. RGB reconstruction uses MSE and the two KL terms in Eq. (6), with $\beta = 2 \times 10^{-4}$; swapping adds no KL term. Grounding uses the RGB kinematic posterior mean through the frozen skeleton decoder; alignment uses paired, ℓ_2-normalized means, with the skeleton target detached. Swap reconstruction uses the frozen skeleton mean and the same residual sample as ordinary RGB reconstruction.

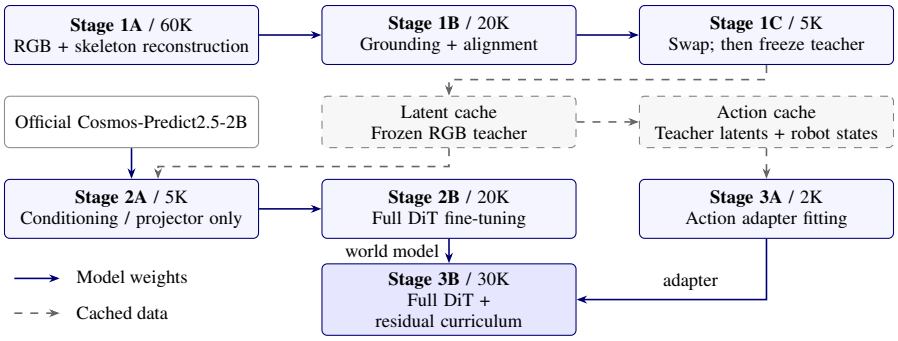


Figure 7: **Training workflow.** Solid arrows transfer model weights; dashed arrows indicate cached data. Stage 3B starts from the Stage 2B world model and Stage 3A adapter and updates all DiT backbone weights. Cache construction does not add optimization stages.

For target skeleton image s, foreground mask m, image prediction $\hat{s}$, and mask prediction $\hat{m}$, the grounding criterion is

$$\begin{aligned} \ell_{\text{ske}} &= \ell_{\text{fg}} + 0.5\ell_{\text{mask}} + 0.05\ell_{\text{bg}}, \\ \ell_{\text{fg}} &= \frac{\sum_{p,c} m_p |\hat{s}_{pc} - s_{pc}|}{C_s \sum_p m_p + \epsilon}, \qquad \ell_{\text{bg}} = \frac{\sum_{p,c} (1-m_p)|\hat{s}_{pc}|}{C_s \sum_p (1-m_p) + \epsilon}, \\ \ell_{\text{mask}} &= \text{BCE}(\hat{m}, m) + 1 - \frac{2\sum_p \hat{m}_p m_p + \epsilon}{\sum_p \hat{m}_p + \sum_p m_p + \epsilon}, \end{aligned} \tag{15}$$

where C_s is the number of color channels. For batch representations Z, the regularizer is

$$\begin{aligned} V_{0.5}(Z) &= \frac{1}{d}\sum_{j=1}^{d} \max(0, 0.5 - \text{Std}_b\, Z_{bj}), \\ \mathcal{L}_{\text{var}} &= \frac{V_{0.5}(Z_{\text{kin}}) + V_{0.5}(Z_{\text{res}})}{2} + \frac{0.1\|\,\text{Cov}(Z_{\text{kin}}, Z_{\text{res}})\|_F^2}{d_{\text{kin}} d_{\text{res}}}. \end{aligned} \tag{16}$$

Cross-covariance uses centered RGB components and excludes the frozen skeleton latent; this discourages linear redundancy, without enforcing independence. Stage 1B minimizes $\mathcal{L}_{\text{rgb}} + \mathcal{L}_{\text{ground}} + 0.5\mathcal{L}_{\text{align}} + 0.1\mathcal{L}_{\text{var}}$. Stage 1C adds $\lambda_{\text{swap}}\mathcal{L}_{\text{swap}}$, with weight 0.5 for 500 updates and 1.0 thereafter; 20% of updates instead use only $\mathcal{L}_{\text{rgb}}$, including its KL terms.

Stages 1B/C use AdamW with $(\beta_1, \beta_2) = (0.9, 0.999)$, $\epsilon = 10^{-8}$, zero weight decay, gradient-norm clipping 0.3, and constant learning rates 5×10^{-5} and 2×10^{-5}, respectively, without warmup. The batch is 32 frame pairs on eight A100 80-GB GPUs (four pairs per GPU); Stage 1C uses bfloat16. At phase transitions only model weights are restored. The configured limits are 40K/10K updates, but the selected checkpoints are 20K/5K.

B.2 World-model training and action adaptation

Stage 2A performs 5K conditioning-projector updates, followed by 20K full-DiT and conditioning updates in Stage 2B. Stage 3A fits only the action adapter for 2K updates with the world model fixed; its learning-rate cycle is configured for 8K updates. Stage 3B starts from these Stage 2B/3A checkpoints and performs 30K action-conditioned generation updates.

Branch dropout. Standardized latents are masked before projection; the carrier contains two 32-dimensional vectors and two binary presence masks (66 values). A dropped branch has zero latent and zero mask. One mode is sampled per clip and shared over all 12 transitions. For modes $(1,1), (0,0), (0,1), (1,0)$, probabilities are $(0.80, 0.10, 0.05, 0.05)$ for global Stage 2 steps 0–999 and $(0.50, 0.10, 0.05, 0.35)$ thereafter. Stage 2B starts at global step 5,000.

Adapter and residual curriculum. The DROID adapter is a per-transition MLP, $7 \to 256 \to 256 \to 32$, with GELU between linear layers. Training-standardized actions predict standardized

RGB kinematic posterior means. With valid-transition indicator v_{ni}, its regression loss is

$$\mathcal{L}_{\text{adapt}} = \frac{\sum_{n,i} v_{ni} \| A_\psi(a_{ni}) - \text{sg}[\mu^{\text{rgb}}_{\text{kin},ni}] \|_2^2}{32 \sum_{n,i} v_{ni} + \epsilon}. \tag{17}$$

In Stage 3B, kinematic conditions always come from actions with mask one. The paired residual is retained with probability 0.5 for updates 0–1,999, decaying linearly to zero over updates 2,000–7,999, and zero thereafter. One choice is shared across a clip; discarded residuals and their masks are zero.

Table 5: **World-model optimization.** Learning rates are module-specific.

Setting	Stage 2B	Stage 3B
Trainable	Full DiT; both conditioning branches	Full DiT; adapter; P_{kin}
Learning rates	DiT: 2.5×10^{-5}; conditioning: 5×10^{-5}	DiT: 2.5×10^{-5}; adapter: 3×10^{-5}; P_{kin}: 10^{-5}
Residual projection	Trainable	Frozen

Both stages use AdamW, $(0.9, 0.999)$, $\epsilon = 10^{-8}$, weight decay 10^{-3}, global gradient clipping 0.1, batch 32, bfloat16, eight-way FSDP, EMA, and 1K checkpoint intervals. Stage 2B uses seed 0 and LambdaLinear scheduling: 100-step warmup, $f_{\text{start}} = 10^{-6}$, $f_{\max} = 0.99$, $f_{\min} = 0.4$, cycle 400K. Normalization statistics remain fixed in Stage 3B.

DROID inference. One observed frame and 12 action transitions generate 12 future frames at 480×640, always with zero residual and mask. We use `FlowUniPCMultistepScheduler`, order 2, solver `bh2`, 35 steps, shift 5, and a non-Karras schedule. Guidance is $f_{\text{cond}} + g(f_{\text{cond}} - f_{\text{uncond}})$ with $g = 0$ (conditional prediction); text is the empty string encoded online, with negative prompts disabled. Results use regular, not EMA, weights at Stage 3B step 30K. Each clip is generated once with seed $20260823 + p$ for zero-based manifest position p; eight ranks partition clips without padding or duplication. All 400 generations completed.

C Evaluation Protocols

C.1 Video and motion metrics

Video fidelity. Canonical clips have 13 RGB frames in $[0, 1]$ at 480×640. Video metrics include the initial conditioning frame; motion metrics use the 12 future frames. MSE and MAE average errors over all channels, pixels, and frames per clip, then over valid clips. PSNR is computed from each clip's MSE e_n before averaging: $N_v^{-1} \sum_n 10 \log_{10}(1/e_n)$ dB. SSIM uses `skimage.metrics.structural_similarity`, `data_range=1.0`, `channel_axis=-1`, and the default 7×7 uniform window. LPIPS uses TorchMetrics AlexNet with `normalize=True`. Both are averaged over frames and then clips. Temporal LPIPS is the mean over clips of

$$\text{tLPIPS}_n = \frac{1}{12} \sum_{t=1}^{12} \left| D_{\text{LPIPS}}(\hat{I}_{n,t}, \hat{I}_{n,t-1}) - D_{\text{LPIPS}}(I_{n,t}, I_{n,t-1}) \right|. \tag{18}$$

It measures agreement in adjacent-frame perceptual change, not end-effector motion.

Validity and displacement. A valid arm trajectory requires in-image reference positions and complete prediction tracking at every evaluated frame, with finite coordinates throughout. Invalid trajectories are excluded from scores but remain in the coverage denominator (400 for DROID, 800 for AgiBot). Thus methods can have different valid subsets; generated-video and valid-video counts are reported separately. For valid trajectories, let $\Delta p_{nt} = p_{nt} - p_{n0}$ and $\Delta \hat{p}_{nt} = \hat{p}_{nt} - \hat{p}_{n0}$. Then $\text{ADE} = (12 N_m)^{-1} \sum_{n=1}^{N_m} \sum_{t=1}^{12} \| \Delta \hat{p}_{nt} - \Delta p_{nt} \|_2$ pixels.

Direction and magnitude. Direction uses initial-to-final displacement: the sign of the horizontal component determines left/right when $|\Delta x| \geq |\Delta y|$; otherwise the vertical sign determines up/down (positive image y is down). Displacements below 0.005 image diagonals (4 pixels) are stationary.

Four-class direction macro-F1 includes only trajectories for which both reference and prediction are nonstationary, a potentially smaller subset than ADE. Magnitude is $m_n = \|\Delta p_{n,12}\|_2/800$; magnitude MAE is $N_m^{-1}\sum_n |\hat{m}_n - m_n|$, and r_{mag} and ρ_{mag} are Pearson and Spearman correlations over valid trajectories. Undefined correlations are unavailable, not zero.

C.2 Baselines, probes, and interventions

Table 1 lists control interfaces; text-only TesserAct (RGB-LoRA) and GE-Base are visual references. Linear probes use frozen encoders and $\widehat{\Delta p} = Wz + b$: displacement MAE/$R^2$ and magnitude macro-F1. Individual latents are 32D, their concatenation 64D; the skeleton teacher uses privileged input.

For interventions, deterministic means k, r, s denote RGB kinematic, RGB residual, and skeleton kinematic latents. Normal, correct/wrong swap, and zero-kinematic/residual inputs are $[k; r]$, $[s; r]$, $[s'; r]$, $[0; r]$, and $[k; 0]$; s' comes from another episode. The decoder, current frame, and target are fixed. Figure 4 reports $100(\bar{e}_j - \bar{e}_0)/\bar{e}_0$ for MAE and $10^3(\overline{\mathrm{SSIM}}_0 - \overline{\mathrm{SSIM}}_j)$. Raw decoder-latent zeroing differs from world-model dropout of standardized latents and presence masks. These tests support motion specialization without establishing complete disentanglement.

C.3 Human transfer and policy evaluation

Human-to-robot transfer. UniSkill (Kim et al., 2025) videos provide RGB-encoder latent sequences directly to the world model, bypassing the action adapter; a robot observation specifies the initial scene. All weights remain fixed, and DreamDojo uses its latent-conditioned checkpoint. The two examples assess visible motion correspondence, not task completion or reliable transfer across arbitrary viewpoints and embodiments.

Open-loop policy evaluation. Seven RoboArena policies are evaluated in 15 sessions (105 paired episodes), with nine real successes and 96 failures. Policy identifiers are `pi0_fast`, `paligemma_fast`, `paligemma_fast_specialist`, `paligemma_vq`, `pi0`, `paligemma_binning`, and `paligemma_diffusion`, each with suffix `_droid`. Models receive the same initial observations and recorded actions; human annotators assign binary task-success labels to generated rollouts using the real-execution criterion. Actions are replayed without policy feedback from generated observations. Episode-level metrics pool all pairs, treating success as positive: success recall, balanced accuracy (mean sensitivity and specificity), and MCC. For policy p, real and generated success rates $r_p, \hat{r}_p$ average the 15 session labels. Pearson and Spearman correlations use seven rate pairs, with average ranks for ties; success-rate MAE is $\Delta_{\mathrm{SR}} = (100/7)\sum_p |r_p - \hat{r}_p|$ percentage points. The seven policies and nine real successes limit the ranking evidence.

D Supplementary Results and Ablations

Additional metrics and coverage. Table 6 complements Table 1 with DROID metrics and trajectory coverage. OSCAR has lower tLPIPS than SkeLAM (0.0174 vs. 0.0222), while SkeLAM has lower ADE and LPIPS. Tables 7 and 8 report AgiBot results: SkeLAM improves SSIM over DreamDojo but has worse LPIPS and ADE, so DROID gains do not extend uniformly to AgiBot. Table 9 compares adaptation budgets.

Component ablations. In Table 2, removing skeleton supervision omits grounding, alignment, and swap, retaining RGB reconstruction; removing swap retains grounding and alignment with $\lambda_{\mathrm{swap}} = 0$. Removing the residual uses zero residual from Stage 2 onward. Removing action adaptation skips adapter regression and jointly trains a randomly initialized adapter and world model for action-conditioned generation. The latter two retain the full model's action interface and inference settings, including zero residual. These change training; decoder interventions keep weights fixed.

Table 6: **DROID: additional metrics and coverage.** All methods generate 400 valid videos. Primary metrics appear in Table 1; N_m counts valid arm trajectories out of 400.

Model	N_m	MAE↓	tLPIPS↓	ρ_{mag} ↑
TesserAct	315/400	0.0928	0.0993	0.0431
GE-Base	330/400	0.1587	0.0549	0.0340
IRASim	328/400	0.1917	0.0912	0.1318
Ctrl-World	326/400	0.0859	0.0220	0.4980
Cosmos-Predict2.5	326/400	0.1873	0.0596	0.0351
Cosmos3	328/400	0.0753	0.0570	0.2080
EnerVerse-AC	327/400	0.1208	0.0584	-0.0448
GE-Sim	328/400	0.1578	0.0534	0.0346
OSCAR	325/400	0.1531	0.0174	0.5030
DreamDojo	325/400	0.0532	0.0366	0.3032
CD-LAM	325/400	0.0571	0.0352	0.2496
SkeLAM	326/400	0.0471	0.0222	0.5283

Table 7: **AgiBot World: video fidelity.** All methods generate 400 videos; N_v is the valid-video count. MSE/MAE use normalized RGB; PSNR is in dB.

Model	N_v	MSE↓	MAE↓	PSNR↑	SSIM↑	LPIPS↓	tLPIPS↓
TesserAct	376	0.0223	0.0684	17.19	0.7781	0.2043	0.0673
GE-Base	400	0.0077	0.0392	22.15	0.7732	0.3004	0.0268
IRASim	400	0.1889	0.3568	7.25	0.1489	0.8636	0.5998
Ctrl-World	376	0.0364	0.1146	14.69	0.5471	0.5313	0.0243
Cosmos-Predict2.5	400	0.0576	0.1627	12.67	0.4610	0.5950	0.0408
Cosmos3	400	0.0151	0.0523	19.74	0.7982	0.1740	0.0315
EnerVerse-AC	376	0.0355	0.1113	14.81	0.5425	0.4484	0.0185
GE-Sim	376	0.0073	0.0383	22.66	0.8459	0.1506	0.0260
OSCAR	376	0.0058	0.0329	23.44	0.8399	0.0966	0.0126
DreamDojo	400	0.0067	0.0341	22.79	0.8330	0.0889	0.0134
CD-LAM	400	0.0092	0.0421	20.97	0.7994	0.1172	0.0113
SkeLAM	400	0.0088	0.0344	21.79	0.8479	0.1049	0.0181

Table 8: **AgiBot World: motion and coverage.** N_m counts valid arm trajectories out of 800. ADE is in pixels and magnitude error is normalized by the image diagonal.

Model	N_m	ADE↓	F1_{dir} ↑	L_1^{mag} ↓	r_{mag} ↑	ρ_{mag} ↑
TesserAct	684/800	45.74	0.2348	0.0850	-0.0489	0.0548
GE-Base	754/800	16.71	0.4830	0.0336	0.1224	0.5104
IRASim	750/800	78.81	0.2628	0.1182	-0.0335	0.0027
Ctrl-World	707/800	18.52	0.2652	0.0382	0.1127	0.0317
Cosmos-Predict2.5	754/800	26.61	0.2802	0.0481	0.0491	0.0816
Cosmos3	752/800	28.61	0.2853	0.0477	0.0972	0.4452
EnerVerse-AC	709/800	18.27	0.2276	0.0376	0.0442	0.1576
GE-Sim	707/800	17.22	0.2529	0.0359	0.0331	0.3021
OSCAR	708/800	17.32	0.5557	0.0309	0.1946	0.6831
DreamDojo	754/800	19.55	0.5720	0.0338	0.2056	0.6299
CD-LAM	750/800	22.73	0.5273	0.0394	0.1583	0.5983
SkeLAM	755/800	21.10	0.2824	0.0366	0.1188	0.5397

Table 9: **Stage 3B adaptation budgets.** Each row uses 400 videos; N_m lists valid/total arm trajectories. Budgets exclude pretraining and the common 2K adapter fit; inference uses zero residual. DROID LPIPS decreases with updates, but motion metrics on both datasets are nonmonotonic.

Data	Steps	N_m	ADE↓	SSIM↑	LPIPS↓	tLPIPS↓	F1_{dir} ↑	r_{mag} ↑
DROID	5K	325/400	45.25	0.8299	0.1374	0.0241	0.5257	0.4930
	10K	326/400	45.37	0.8326	0.1337	0.0232	0.5248	0.4986
	15K	327/400	42.48	0.8345	0.1314	0.0228	0.5439	0.5408
	30K	326/400	41.15	0.8340	0.1310	0.0222	0.5465	0.5277
AgiBot	5K	756/800	20.79	0.8452	0.1073	0.0185	0.2968	0.1742
	10K	754/800	21.29	0.8477	0.1056	0.0182	0.2888	0.1332
	15K	755/800	21.14	0.8471	0.1057	0.0182	0.2752	0.1639
	30K	755/800	21.10	0.8479	0.1049	0.0181	0.2824	0.1188

E Additional Qualitative Comparisons

Figures 8 and 9 supplement Figure 3 with two DROID scenes. Both grids compare recorded observations and generated rollouts at four time steps, showing end-effector motion, arm geometry, and scene appearance.

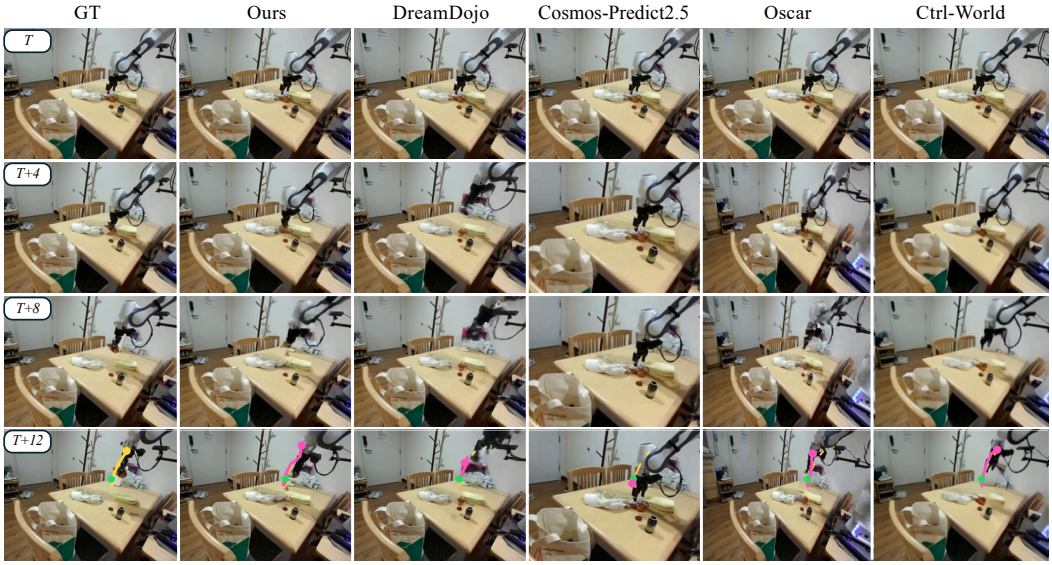


Figure 8: **Tabletop scene.** Columns: ground truth, SkeLAM, DreamDojo, Cosmos-Predict2.5, OSCAR, and Ctrl-World. Rows: T, $T+4$, $T+8$, and $T+12$. Colored markers and arrows highlight end-effector motion in the final row.

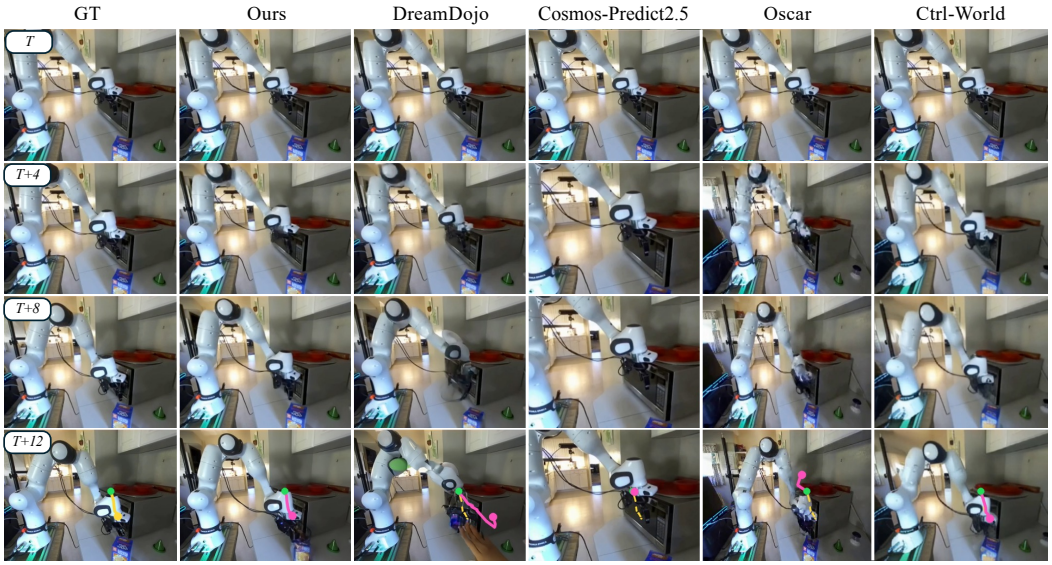


Figure 9: **Kitchen scene.** Models and time steps follow Figure 8; markers and arrows highlight end-effector motion in the final row.

AI use statement. Generative AI assisted language polishing and readability; all AI-assisted content was reviewed by authors.